

Activity 3 - Share inventory records with ownership and borrowing

AI Vibe Coding with Rust / TGS-2023039924 / version 3.0 / K1/A1 K3/A3

Specification

Item { sku, name, qty: u32 }. total_quantity(&[Item]) -> u64 reads; restock(&mut Item, u32) -> Option<u32> changes one item and leaves it unchanged on overflow; in_stock(Vec<Item>) -> Vec<Item> consumes the list.

Plan before code

Act as a senior Rust reviewer. Specification: Item { sku, name, qty: u32 }. total_quantity(&[Item]) -> u64 reads; restock(&mut Item, u32) -> Option<u32> changes one item and leaves it unchanged on overflow; in_stock(Vec<Item>) -> Vec<Item> consumes the list. Before writing any code, list the public signatures, the edge cases and at least four boundary tests you would expect. Ask me about anything ambiguous.

Generate with AI

Implement "Share inventory records with ownership and borrowing" in starter/src so that the supplied tests pass. Choose &, &mut or by-value parameters to match each function's intent; avoid clone() unless you justify it. Keep public signatures unchanged, use no unsafe code, and explain each design decision in one sentence.

Repair from evidence

Here is my exact cargo output: <paste the first error or failing test>. Explain the cause in plain English, propose the smallest fix that keeps the specification, and add one regression test. Do not change expected values in existing tests.

Review checklist

Signatures unchanged; no unsafe; no unwrap() on input; no new crates; every boundary tested; cargo test, clippy -D warnings and fmt --check pass; you can explain every line.